



Figure 1: The official portal of Apache Mahout

cluster evaluation, anomaly detection, semi-supervised learning, reinforcement learning and many others.

Free and open source tools for machine learning are Apache Mahout, Scikit-Learn, OpenAI, TensorFlow, Char-RNN, PaddlePaddle, CNTX, Apache Singa, DeepLearning4J, H2O, etc.

Apache Mahout, a scalable high performance machine learning framework

Apache Mahout (mahout.apache.org) is a powerful and high performance machine learning framework for the implementation of machine learning algorithms. It is traditionally used to integrate supervised machine learning algorithms with the target value assigned to each input data set. Apache Mahout can be used for assorted research based applications including social media extraction and sentiment mining, user belief analytics, YouTube analytics and many related real-time applications.

In Apache Mahout, a 'mahout' refers to whatever drives or operates the elephant. The mahout acts as the master of the elephant in association with Apache Hadoop and is represented in the logo of the elephant. Apache Mahout runs with the base installation of Apache Hadoop, and then the machine learning algorithms are implemented with the features to develop and deploy scalable machine learning algorithms. The prime approaches, like recommender engines, classification problems and clustering, can be effectively solved using Mahout.

Corporate users of Mahout include Adobe, Facebook, LinkedIn, FourSquare, Twitter and Yahoo.

Installing Apache Mahout

To start with the Mahout installation, Apache Hadoop has to be set up on a Linux distribution. To get ready with Hadoop, the installation is required to be updated as follows, in Ubuntu Linux:

```
$ sudo apt-get update
$ sudo addgroup hadoop
$ sudo adduser --ingroup hadoop hadoopuser1
$ sudo adduser hadoopuser1 sudo
$ sudo apt-get install ssh
$ su hadoopuser1
$ ssh-keygen -t rsa
$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
$ chmod 0600 ~/.ssh/authorized_keys
$ ssh localhost
```

Installing the latest version of Hadoop

Use the following code to install the latest version of Hadoop:

```
$ wget http://www-us.apache.org/dist/hadoop/common/hadoop-
HadoopVersion/hadoop-HadoopVersion.tar.gz
$ tar xvfz hadoop-HadoopVersion.tar.gz
$ sudo mkdir -p /usr/local/hadoop
$ cd hadoop-HadoopVersion/
$ sudo mv * /usr/local/hadoop
$ sudo chown -R hadoopuser1:hadoop /usr/local/hadoop
$ hadoop namenode -format
$ cd /usr/local/hadoop/sbin
$ start-all.sh
```

The following files are required to be updated next:

- `~/bashrc`
- `core-site.xml`
- `hadoop-env.sh`
- `hdfs-site.xml`
- `mapred-site.xml`
- `yarn-site.xml`

Web interfaces of Hadoop

Listed below are some of the Web interfaces of Hadoop.

MapReduce: <http://localhost:8042/>

NameNode daemon: <http://localhost:50070/>

Resource Manager: <http://localhost:8088/>

SecondaryNameNode: <http://localhost:50090/status.html>

The default port to access Hadoop is 50070 and <http://localhost:50070/> is used on a Web browser.

After installing Hadoop, the setting up of Mahout requires the following code:

```
$ wget http://mirror.nexcess.net/apache/mahout/0.9/mahout-
Distribution.tar.gz
$ tar xvfz mahout-Distribution.tar.gz
```

Implementing the recommender engine algorithm

Nowadays, when we shop at online platforms like Amazon, eBay, SnapDeal, FlipKart and many others, we notice that most of these online shopping platforms give us suggestions or recommendations about the products that we like or had purchased earlier. This type of implementation or